CAP 6315 - Social Networks and Big Data Analytics

Project Part #2

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Colab Project Link

GitHub Project Link

Overleaf Report

This project with be using the *Twitch Social Network* datasets, and be examining the Brazilian Portuguese language stream for our network.

In [30]:
```
# Setup and Imports
import random
import os
import zipfile

import matplotlib.pyplot as plt
import networkx as nx
import numpy as np
from pyspark.ml.classification import (
    LogisticRegression,
    RandomForestClassifier,
)
from pyspark.ml.evaluation import (
    BinaryClassificationEvaluator,
    MulticlassClassificationEvaluator,
)
from pyspark.ml.feature import (
    VectorAssembler,
)
from pyspark.ml.tuning import (
    CrossValidator,
)
from pyspark.sql import (
    SparkSession,
    Window,
)
import pyspark.sql.functions as F
import pyspark.sql.types as T
import seaborn as sns


SEED = 1234
LANG = "PTBR"
PATH = os.getcwd()
DATA_PATH = os.path.join(
    PATH,
    "twitch",
    LANG,
)
```

```
random.seed(SEED)
np.random.seed(SEED)
```

Importing the data and creating PySpark session

Unzipping archive file

In [31]:
```
arhive_file = "twitch.zip"
with zipfile.ZipFile(os.path.join(PATH, arhive_file), "r") as archive_obj:
  archive_obj.extractall(PATH)
```

Creating the Spark Session and Loading the Dataset

In [32]:
```
spark = SparkSession.builder \
    .appName("ProjectPart-2") \
    .getOrCreate()
```

In [33]:
```
df = spark.read.csv(
    path=os.path.join(DATA_PATH, f"musae_{LANG}_edges.csv"),
    header=True,
    inferSchema=True,
)
```

Creating Graph with NetworkX

In [34]:
```
G = nx.Graph()

edges = df.collect()
G.add_edges_from([
    (row["from"], row["to"])
    for row in edges
])
```

Create a Spark Dataframe with the Following Network Features:

- Degree Centrality
- Betweenness Centrality
- Closeness Centrality
- Eigenvector Centrality
- PageRank

In [35]:
```
degree = nx.degree_centrality(G)
betweenness = nx.betweenness_centrality(G)
closeness = nx.closeness_centrality(G)
eigenvector = nx.eigenvector_centrality(G)
pagerank = nx.pagerank(G)
```

Test to Make Sure All Measures are in the Same Order

In [36]:
```
def check_order(dict_1, dict_2):
    """Check if two dictionaries have the same keys in the same order.

    :param dict_1: First dictionary to check
    :param dict_2: Second dictionary to check
```

```
    :return: True if the dictionaries have the same keys in the same order
    """
    return list(dict_1.keys()) == list(dict_2.keys())

print(
    f"Is same order for Degree and Betweenness Centralities:"
    + " "
    f"{check_order(degree, betweenness)}")
print(
    f"Is same order for Degree and Closeness Centralities:"
    + " "
    f"{check_order(degree, closeness)}")
print(
    f"Is same order for Degree and Eigenvector Centralities:"
    + " "
    f"{check_order(degree, eigenvector)}")
print(
    f"Is same order for Degree Centrality and PageRank:"
    + " "
    f"{check_order(degree, pagerank)}")
```

```
Is same order for Degree and Betweenness Centralities: True
Is same order for Degree and Closeness Centralities: True
Is same order for Degree and Eigenvector Centralities: True
Is same order for Degree Centrality and PageRank: True
```

Create Graph Centrality Spark Dataframe

In [37]:

```
schema = T.StructType([
    T.StructField("node", T.IntegerType(), True),
    T.StructField("degree", T.DoubleType(), True),
    T.StructField("betweenness", T.DoubleType(), True),
    T.StructField("closeness", T.DoubleType(), True),
    T.StructField("eigenvector", T.DoubleType(), True),
    T.StructField("pagerank", T.DoubleType(), True),
])

data = zip(
    degree.keys(),
    degree.values(),
    betweenness.values(),
    closeness.values(),
    eigenvector.values(),
    pagerank.values(),
)

g_df = spark.createDataFrame(
    data,
    schema=schema,
)

g_df.show(5)
```

```
+----+--------------------+--------------------+-------------------+--------------------
-+--------------------+
|node|              degree|         betweenness|          closeness|         eigenvecto
r|            pagerank|
+----+--------------------+--------------------+-------------------+--------------------
-+--------------------+
|   0|0.001046572475143...|2.039880322525145...| 0.3489136388533869| 0.0012243497715967
5|1.080941325389477...|
|  92|0.023547880690737835|0.003121403530497...| 0.4241011984021305|0.00861220473914065
3|9.339108322930132E-4|
| 428| 0.17373103087388803|0.021980008718637752| 0.5219885277246654| 0.1127115602630264
6|0.004689928434443929|
|   1|0.001569858712715...|  4.6043920125031E-7|0.35025659824046923|0.00155347620651
8...|1.148048466908686...|
| 689| 0.07430664573521717|0.002383681268152517|0.47870741482965934|0.06106692177070515
6|0.001896727185355...|
+----+--------------------+--------------------+-------------------+--------------------
-+--------------------+
only showing top 5 rows
```

Join Node Attributes with the Network Features

Node Attributes

- Views
- Days
- Mature

Target Class

- Partner

Load Node Attribute Dataframe

In [38]:
```
att_df = spark.read.csv(
    path=os.path.join(DATA_PATH, f"musae_{LANG}_target.csv"),
    header=True,
    inferSchema=True
)

att_df.show(5)
```

```
+---------+----+------+------+-------+------+
|       id|days|mature| views|partner|new_id|
+---------+----+------+------+-------+------+
| 44891403|1943| false|929459|   true|  1706|
| 61180621|1633| false| 11194|  false|  1273|
|145109685| 632| false|  2850|  false|   313|
|122121521| 906|  true|  3422|  false|  1570|
|189445819| 267| false|    71|  false|   800|
+---------+----+------+------+-------+------+
only showing top 5 rows
```

Join Graph Attributes and Node Attributes Dataframes

In [39]:
```
join_df = g_df.join(
    other=att_df,
    on=[g_df.node == att_df.new_id],
    how="inner",
)
```

```
join_df = join_df.drop("id", "new_id")

join_df.show(5)
```

```
+----+--------------------+--------------------+--------------------+--------------------
-+--------------------+----+------+-------+-------+
|node|              degree|         betweenness|          closeness|         eigenvecto
r|            pagerank|days|mature|  views|partner|
+----+--------------------+--------------------+--------------------+--------------------
-+--------------------+----+------+-------+-------+
|   0|0.001046572475143...|2.039880322525145...| 0.3489136388533869| 0.0012243497715967
5|1.080941325389477...|1601| false|    475|  false|
|  92|0.023547880690737835|0.003121403530497...| 0.4241011984021305|0.00861220473914065
3|9.339108322930132E-4|2003|  true|  60988|   true|
| 428| 0.17373103087388803|0.021980008718637752| 0.5219885277246654| 0.1127115602630264
6|0.004689928434443929| 845|  true| 552640|   true|
|   1|0.001569858712715...|  4.6043920125031E-7|0.35025659824046923|0.00155347620651
8...|1.148048466908686...|1561| false|    345|  false|
| 689| 0.07430664573521717|0.002383681268152517|0.47870741482965934|0.06106692177070515
6|0.001896727185355...|2295|  true|1796373|   true|
+----+--------------------+--------------------+--------------------+--------------------
-+--------------------+----+------+-------+-------+
only showing top 5 rows
```

Define the Target Variable as *partner*

NOTE: Since the only categorical feature *mature* is a boolean type the *StringIndexer* or *OneHotEncoder* was NOT needed for creating the features.

In [40]:

```
target_column = "partner"

feature_columns = [
    "degree",
    "betweenness",
    "closeness",
    "eigenvector",
    "pagerank",
    "days",
    "mature",
    "views",
]

assembler = VectorAssembler(
    inputCols=feature_columns,
    outputCol="features",
)
feature_df = assembler.transform(join_df)

feature_df.select("features", target_column).show(5, truncate=False)
```

```
+-----------------------------------------------------------------------------------------
----------------------------------------+-------+
|features
|partner|
+-----------------------------------------------------------------------------------------
----------------------------------------+-------+
|[0.0010465724751439038,2.0398803225251453E-6,0.3489136388533869,0.00122434977159675,1.
0809413253894772E-4,1601.0,0.0,475.0]   |false  |
|[0.023547880690737835,0.0031214035304972156,0.4241011984021305,0.008612204739140653,9.
339108322930132E-4,2003.0,1.0,60988.0]  |true   |
|[0.17373103087388803,0.021980008718637752,0.5219885277246654,0.11271156026302646,0.004
689928434443929,845.0,1.0,552640.0]     |true   |
|[0.0015698587127158557,4.6043920125031E-7,0.35025659824046923,0.0015534762065188574,1.
1480484669086864E-4,1561.0,0.0,345.0]   |false  |
|[0.07430664573521717,0.002383681268152517,0.47870741482965934,0.061066921770705156,0.0
018967271853552571,2295.0,1.0,1796373.0]|true   |
+-----------------------------------------------------------------------------------------
----------------------------------------+-------+
only showing top 5 rows
```

Calculate the Total Number of Partners and Non-Partners

In [41]:
```python
print(
    f"Non-Partnered Count: {feature_df.filter(~F.col(target_column)).count()}")
print(f"Partnered Count: {feature_df.filter(F.col(target_column)).count()}")
```

```
Non-Partnered Count: 1633
Partnered Count: 279
```

Partner and Non-Partner Dataset Balance

> The dataset is comprised of $1,633$ non-partnered streamer which makes up around 85.4% of the data. So, there is a large imbalance in the dataset for the partner class that favors the non-partnered streamers.

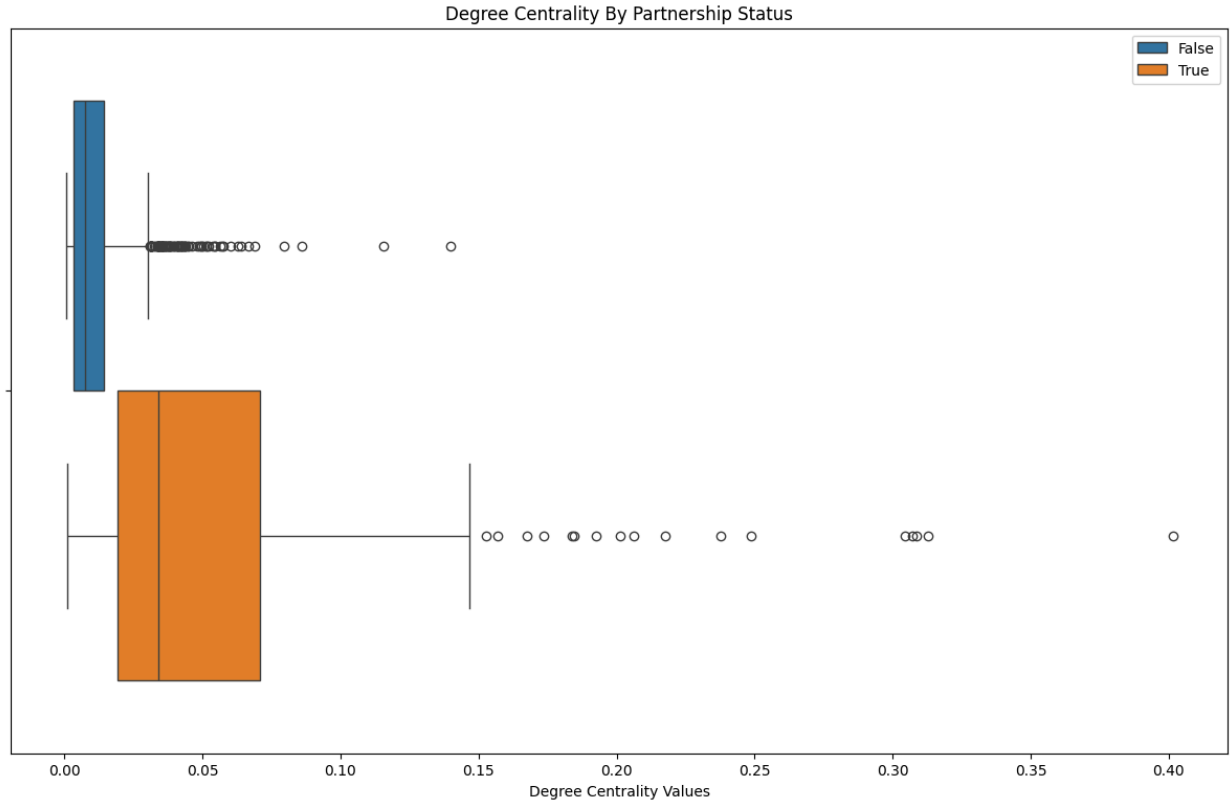
Numerical Feature Boxplots

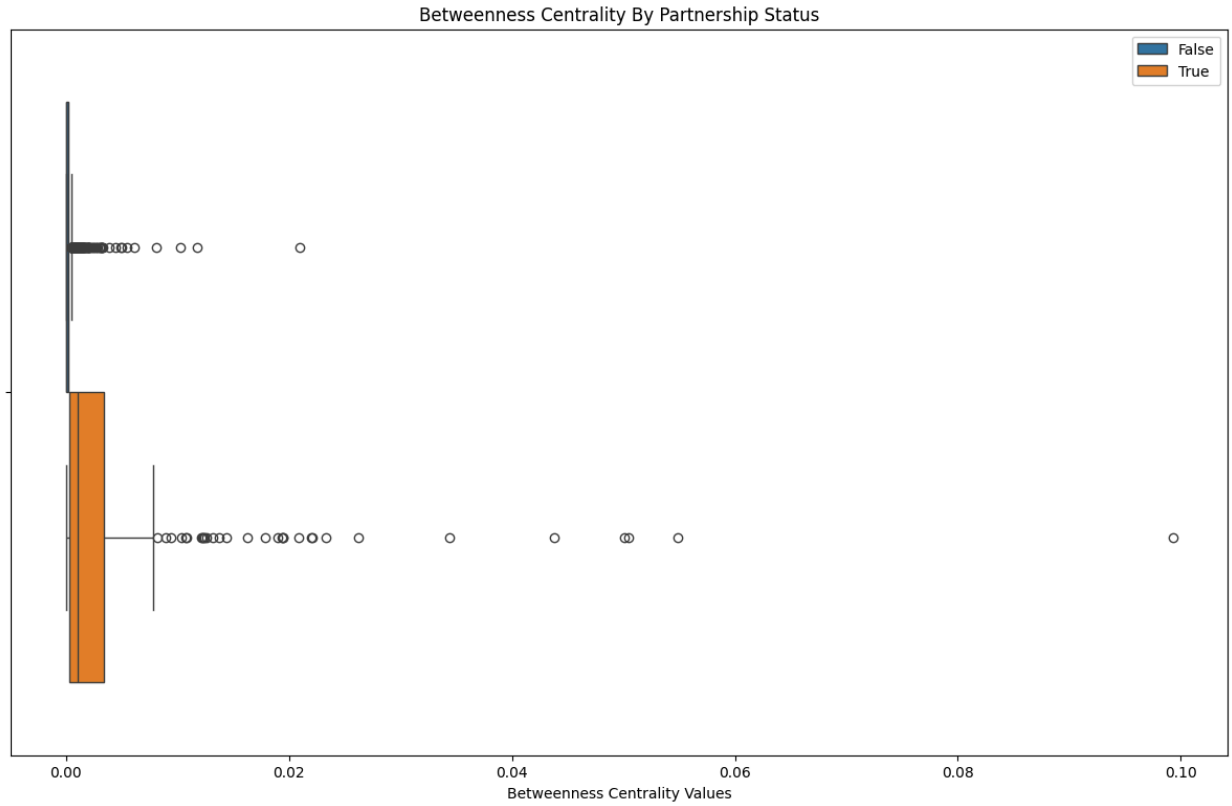
The Boxplots for the Numerical Features are Grouped by Partner Status

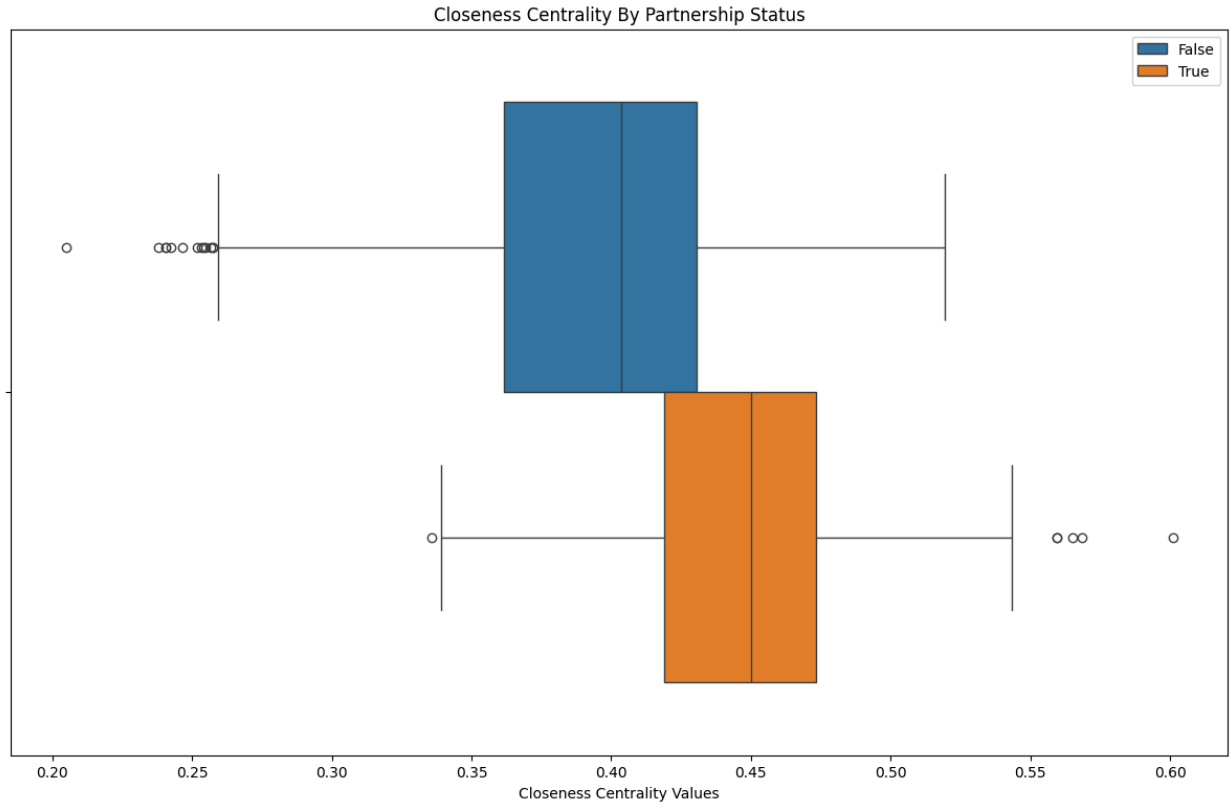
In [42]:
```python
def numeric_boxplots(df, feature, target, title, x_label):
    """Shows the boxplot for a numerical feature grouped by the target column.

    :param df: PySpark Dataframe
    :param feature: Name of the feature column
    :param target: Name of the target column
    :param title: Title of the plot
    :param x_label: Label for the x-axis
    """
    plt.figure(figsize=(15, 9))
    sns.boxplot(
        x=df.select(feature).rdd.map(lambda x: x[0]).collect(),
        hue=df.select(target).rdd.map(lambda x: x[0]).collect(),
    )
    plt.title(title)
    plt.xlabel(x_label)
    plt.show()
```

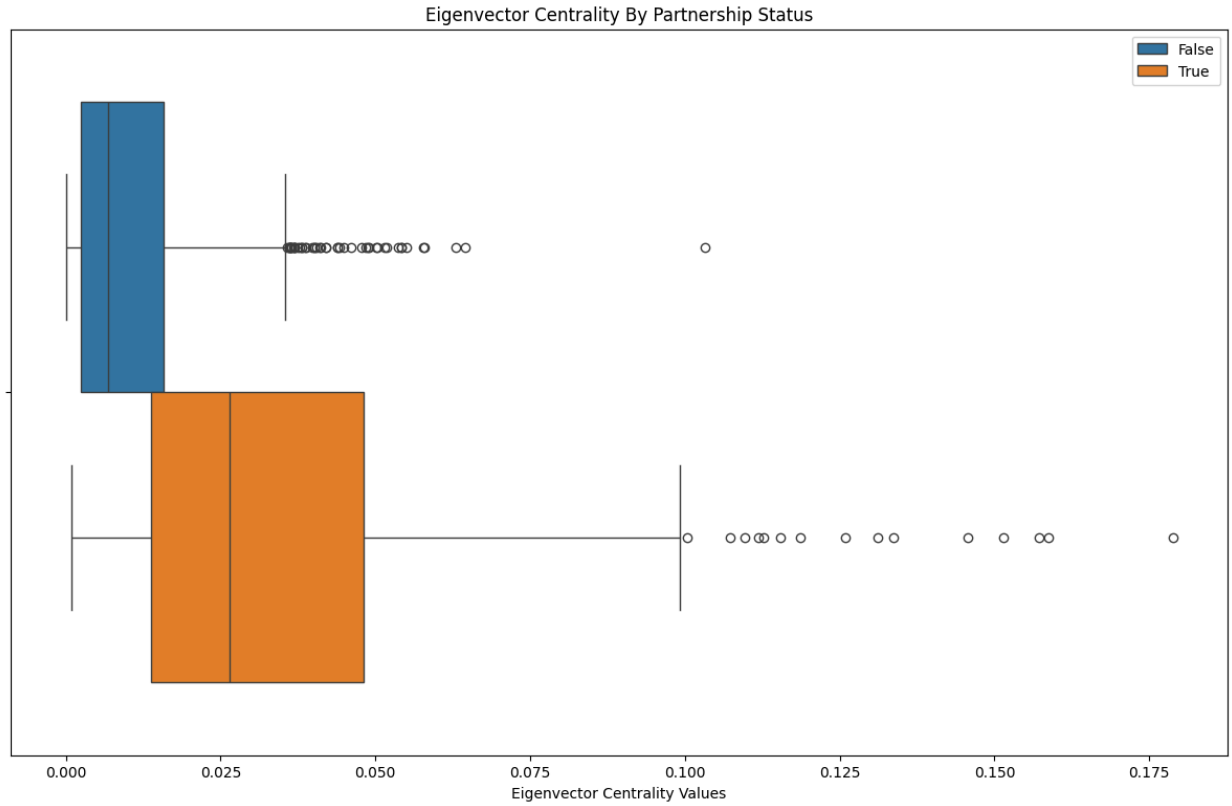
In [43]:
```python
cent_boxplots = feature_columns[:4]
```

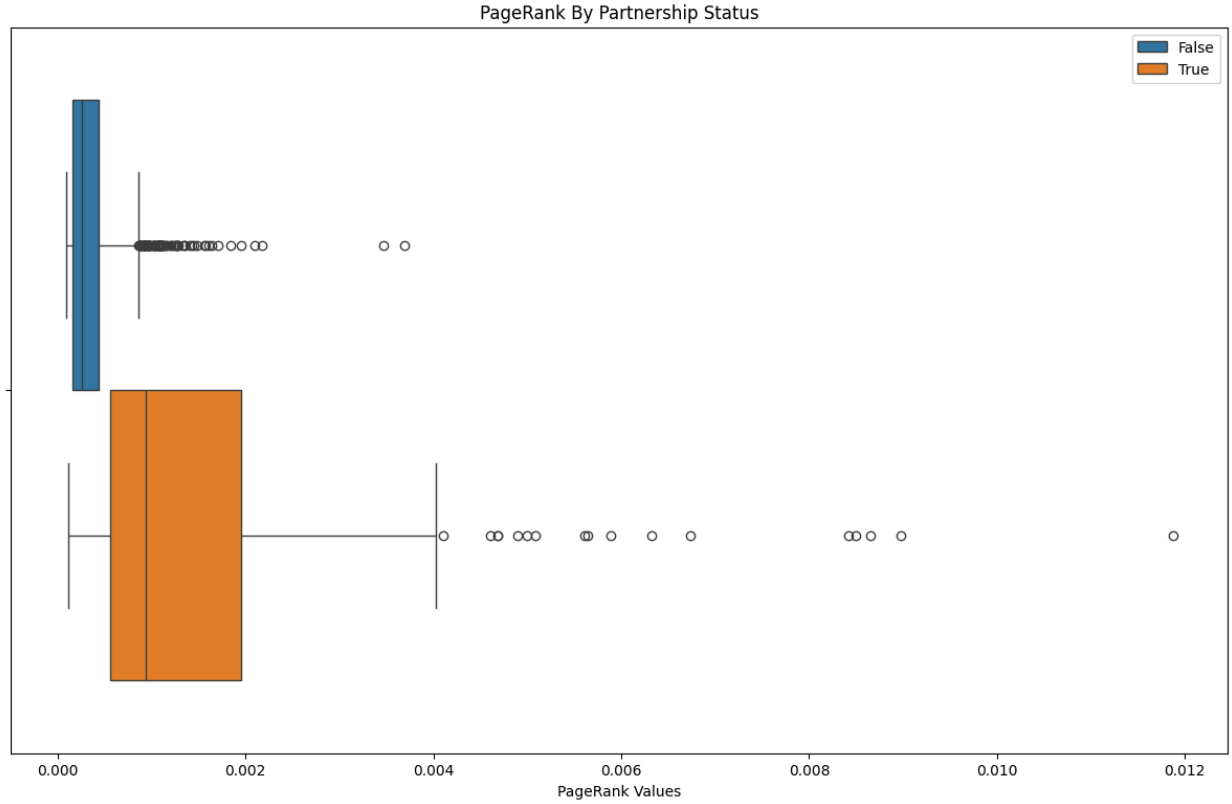
```python
for feature in cent_boxplots:
    numeric_boxplots(
        df=feature_df,
        feature=feature,
        target=target_column,
        title=f"{feature.title()} Centrality By Partnership Status",
        x_label=f"{feature.title()} Centrality Values",
    )
    # plt.figure(figsize=(15, 9))
    # sns.boxplot(
    #     x=feature_df.select(feature).rdd.map(lambda x: x[0]).collect(),
    #     hue=feature_df.select("partner").rdd.map(lambda x: x[0]).collect(),
    # )
    # plt.title(f"{feature.title()} Centrality By Partnership Status")
    # plt.xlabel(f"{feature.title()} Centrality Values")
    # plt.show()

num_boxplots = [
    "pagerank",
    "days",
    "views",
]

for feature in num_boxplots:
    name = "PageRank" if feature == "pagerank" else feature.title()
    x_label = f"{name} Values" if feature == "pagerank" else name
    numeric_boxplots(
        df=feature_df,
        feature=feature,
        target=target_column,
        title=f"{name} By Partnership Status",
        x_label=x_label,
    )
    # plt.figure(figsize=(15, 9))
    # sns.boxplot(
    #     x=feature_df.select(feature).rdd.map(lambda x: x[0]).collect(),
    #     hue=feature_df.select("partner").rdd.map(lambda x: x[0]).collect(),
    # )
    # name = "PageRank" if feature == "pagerank" else feature.title()
    # x_label = f"{name} Values" if feature == "pagerank" else name
    # plt.title(f"{name} By Partnership Status")
    # plt.xlabel(x_label)
    # plt.show()
```

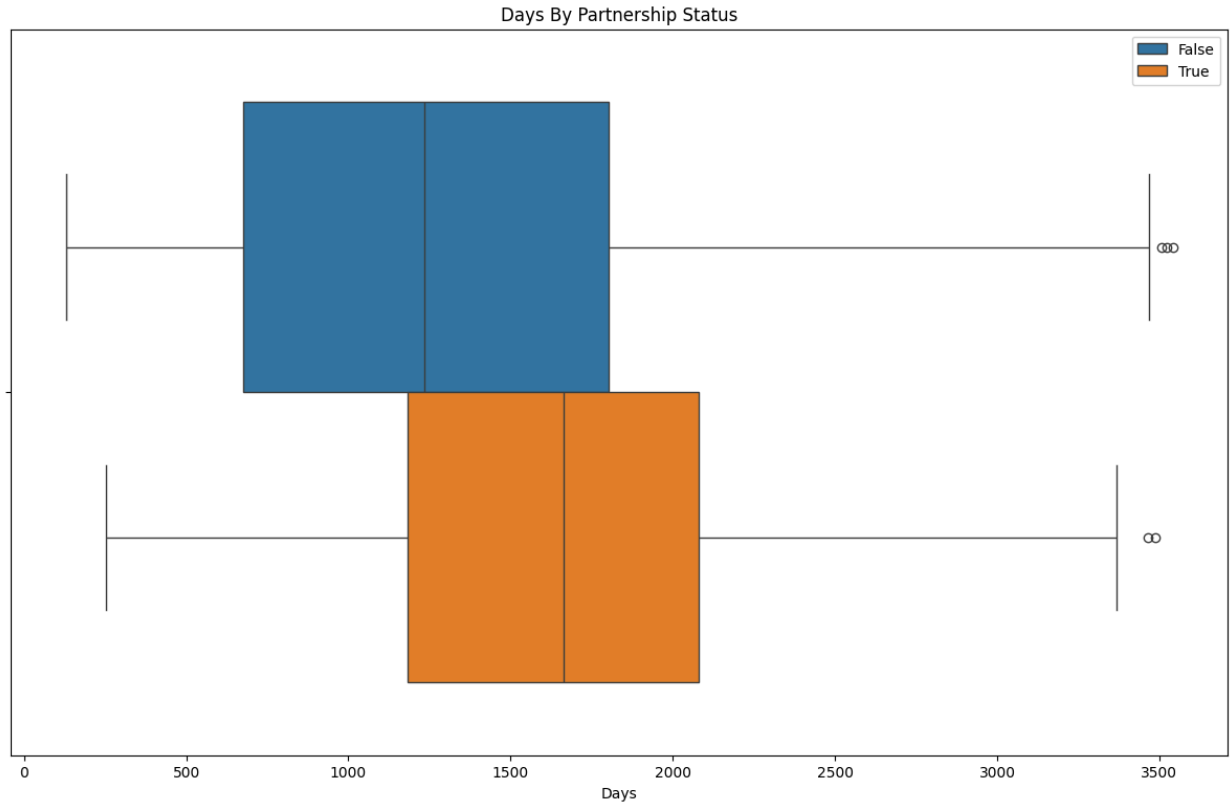

Degree Centrality By Partnership Status
False
True
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
Degree Centrality Values

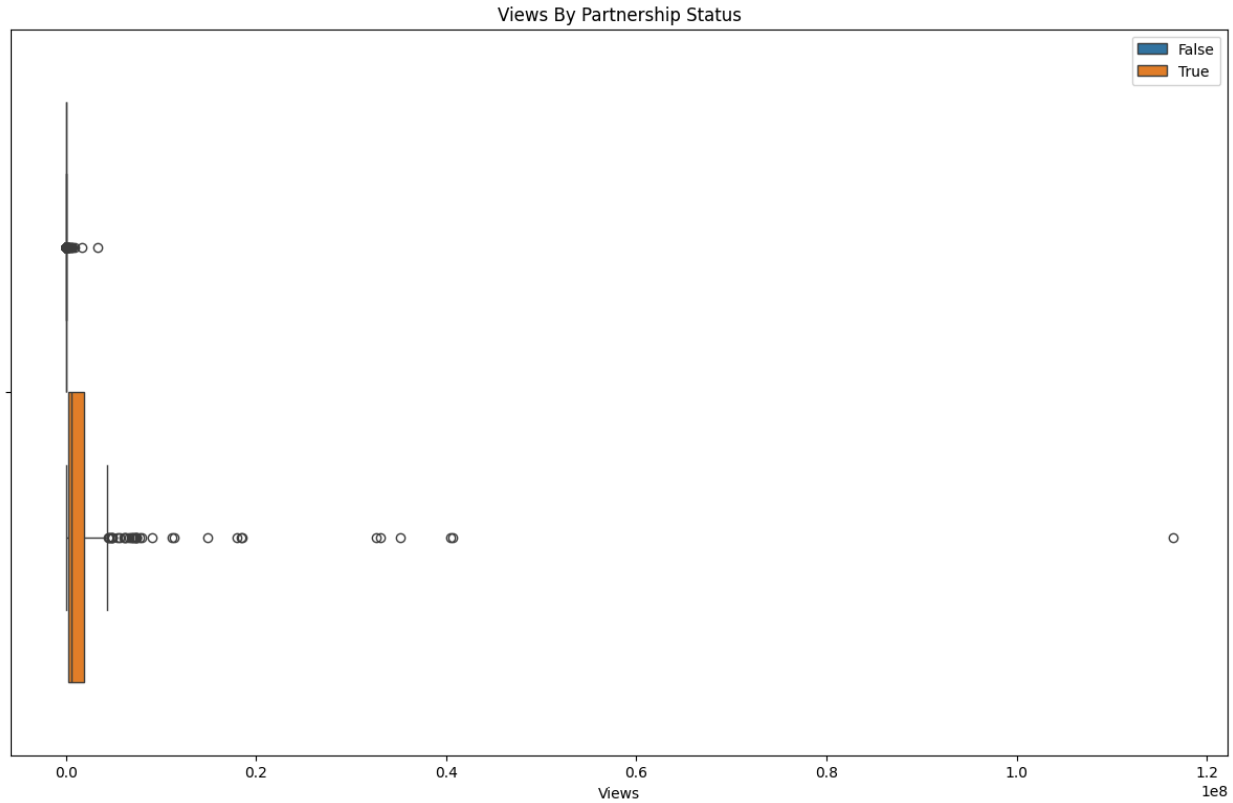

Betweenness Centrality By Partnership Status
False
True
0.00
0.02
0.04
0.06
0.08
0.10
Betweenness Centrality Values


Closeness Centrality By Partnership Status
False
True
0.20
0.25
0.30
0.35
0.40
0.45
0.50
0.55
0.60
Closeness Centrality Values


Eigenvector Centrality By Partnership Status
False
True
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Eigenvector Centrality Values


PageRank By Partnership Status
False
True
0.000
0.002
0.004
0.006
0.008
0.010
0.012
PageRank Values


Days By Partnership Status
False
True
0
500
1000
1500
2000
2500
3000
3500
Days



Categorical Features Bar Chart

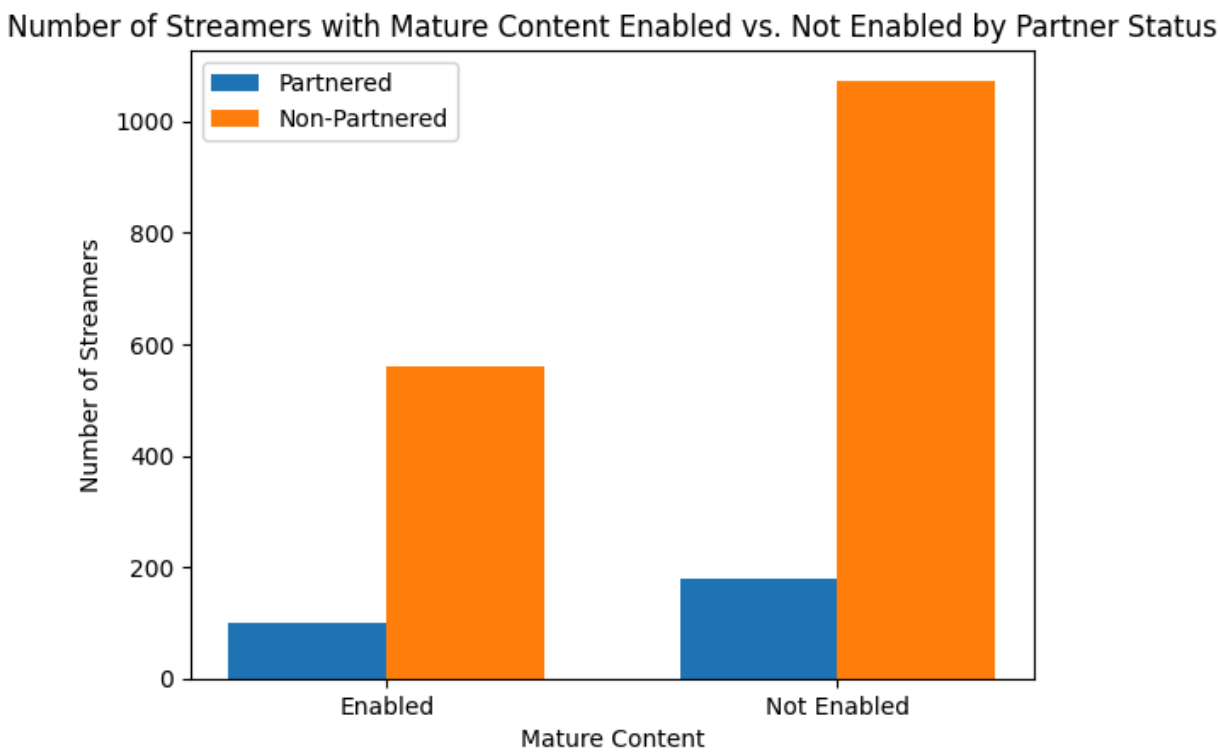
The Bar Charts for the Categorical Features are Grouped by Partner Status

In [44]:
```
feature_df.groupBy("mature", "partner").count().show()

x_label = ["Enabled", "Not Enabled"]
x_ind = np.arange(len(x_label))
y_1 = feature_df.filter(F.col(target_column)) \
    .groupBy("mature").count().rdd.map(lambda x: x[1]).collect()
y_2 = feature_df.filter(~F.col(target_column)) \
    .groupBy("mature").count().rdd.map(lambda x: x[1]).collect()
width = 0.35

fig, ax = plt.subplots()
ax.bar(
    x=x_ind - width / 2,
    height=y_1,
    width=width,
    label="Partnered",
)
ax.bar(
    x=x_ind + width / 2,
    height=y_2,
    width=width,
    label="Non-Partnered",
)
ax.set_ylabel("Number of Streamers")
ax.set_title(
    "Number of Streamers with Mature Content Enabled vs. Not Enabled by Partner"
    + " "
    + "Status")
ax.set_xticks(x_ind)
ax.set_xticklabels(x_label)
```

```
ax.set_xlabel("Mature Content")
ax.legend()
```

```
+------+-------+-----+
|mature|partner|count|
+------+-------+-----+
|  true|  false|  560|
|  true|   true|  101|
| false|  false| 1073|
| false|   true|  178|
+------+-------+-----+
```

Out[44]: <matplotlib.legend.Legend at 0x79791e42b830>



Feature Discussion on Visualizations

> Based on the visualizations, the features that have a visible difference where mean ($Q2$) of one group is above ($Q3$) or below ($Q1$) of the other group for numerical features or double the streamers of the other group for categorical features between partner and non-partner streamers are:
>
> - Degree Centrality
> - Betweenness Centrality
> - Closeness Centrality
> - Eigenvector Centrality
> - PageRank
> - Views
> - Mature Content
>
> The numerical feature *Days* was the only feature that did not show a massive difference that we were looking for above.

Note: This is before the data imbalanced is handled.

Handling the Class Imbalance

In [45]:
```
minor_df = feature_df.filter(F.col(target_column))
major_df = feature_df.filter(~F.col(target_column))

major_sampled = major_df.sample(
    fraction=minor_df.count() / major_df.count(),
    seed=SEED,
)

print(f"Minority Class Count: {minor_df.count()}")
print(f"Majority Sampled Count: {major_sampled.count()}")

balanced_df = minor_df.union(major_sampled)
balanced_df.groupBy(target_column).count().show()
```

```
Minority Class Count: 279
Majority Sampled Count: 294
+-------+-----+
|partner|count|
+-------+-----+
|   true|  279|
|  false|  294|
+-------+-----+
```

Split the Data into Training and Testing Sets

- This experiment is using an 80% Training to 20% Testing Split
- The Target Column Needed to be Converted to a Numerical Value

In [46]:
```
shuffled_df = balanced_df \
    .withColumn("partner", F.col("partner").cast(T.IntegerType())) \
    .orderBy(F.rand(seed=SEED))

shuffled_df.show(5)
```

```
+----+--------------------+--------------------+-------------------+--------------------
-+--------------------+----+------+------+-------+--------------------+
|node|              degree|         betweenness|          closeness|         eigenvecto
r|            pagerank|days|mature| views|partner|            features|
+----+--------------------+--------------------+-------------------+--------------------
-+--------------------+----+------+------+-------+--------------------+
| 887|0.016221873364730507|0.001124225028057851| 0.4246666666666667|0.01338989579134259
4|5.468729589767586E-4| 751|  true|560606|      0|[0.01622187336473...|
|  21|0.012035583464154894|5.343165711693281E-5| 0.4169757800567314|0.01222169642528425
8|3.523741879322434...|1451|  true|  8728|      0|[0.01203558346415...|
| 345|0.004709576138147567| 4.21831282408123E-5|0.37640338782745714|0.00302881508886
8...|2.140813055632411E-4|1661|  true|  6938|      0|[0.00470957613814...|
|1418|0.004709576138147567|1.116215687569845...| 0.4119422289286484|0.00951273501011654
4|1.871424995014235...| 370|  true|  2301|      0|[0.00470957613814...|
|1506|0.025641025641025644|2.625531560257615E-4| 0.3602262016965127|0.00324990428782
6...|7.982181871439684E-4|3507| false| 10911|      0|[0.02564102564102...|
+----+--------------------+--------------------+-------------------+--------------------
-+--------------------+----+------+------+-------+--------------------+
only showing top 5 rows
```

In [47]:
```
train_df, test_df = shuffled_df.randomSplit(
    weights=[0.8, 0.2],
    seed=SEED,
```

```
)
print(f"Training Set Size: {train_df.count()}")
print(f"Testing Set Size: {test_df.count()}")
```

```
Training Set Size: 453
Testing Set Size: 120
```

Train a Logistic Regression Model

Using PySpark Perform a **5-fold** Cross-Validation

In [48]:

```
k = 5
folds = train_df.randomSplit(
    weights=[1 / k] * k,
    seed=SEED,
)

acc_scores = []
f1_scores  = []
precision_scores = []
recall_scores = []

for i in range(k):
    test_fold = folds[i]
    train_fold = spark.createDataFrame(
        spark.sparkContext.emptyRDD(),
        train_df.schema,
    )

    for j in range(k):
        if j != i:
            train_fold = train_fold.union(folds[j])

    lr_mod = LogisticRegression(
        featuresCol="features",
        labelCol=target_column,
    ).fit(train_fold)
    preds = lr_mod.transform(test_fold)

    acc_evaluator = MulticlassClassificationEvaluator(
        labelCol=target_column,
        predictionCol="prediction",
        metricName="accuracy",
    )
    f1_evaluator = MulticlassClassificationEvaluator(
        labelCol=target_column,
        predictionCol="prediction",
        metricName="f1",
    )
    precision_evaluator = MulticlassClassificationEvaluator(
        labelCol=target_column,
        predictionCol="prediction",
        metricName="precisionByLabel",
    )
    recall_evaluator = MulticlassClassificationEvaluator(
        labelCol=target_column,
        predictionCol="prediction",
        metricName="recallByLabel",
    )

    acc_scores.append(acc_evaluator.evaluate(preds))
    f1_scores.append(f1_evaluator.evaluate(preds))
```

```
    precision_scores.append(precision_evaluator.evaluate(preds))
    recall_scores.append(recall_evaluator.evaluate(preds))
```

Report the Average Cross-Validation Metrics

- Accuracy
- F1 Score
- Precision
- Recall

In [49]:
```
print(f"Average Accuracy: {sum(acc_scores) / k}")
print(f"Average F1 Score: {sum(f1_scores) / k}")
print(f"Average Precision: {sum(precision_scores) / k}")
print(f"Average Recall: {sum(recall_scores) / k}")
```

```
Average Accuracy: 0.9034604893263947
Average F1 Score: 0.9032815755432628
Average Precision: 0.8770457090501884
Average Recall: 0.9457367213402718
```

Train the Model

Train Model on the Whole Training Set and Generate Predictions on the Test Set

In [50]:
```
acc_evaluator = MulticlassClassificationEvaluator(
    labelCol=target_column,
    predictionCol="prediction",
    metricName="accuracy",
)
f1_evaluator = MulticlassClassificationEvaluator(
    labelCol=target_column,
    predictionCol="prediction",
    metricName="f1",
)
precision_evaluator = MulticlassClassificationEvaluator(
    labelCol=target_column,
    predictionCol="prediction",
    metricName="precisionByLabel",
)
recall_evaluator = MulticlassClassificationEvaluator(
    labelCol=target_column,
    predictionCol="prediction",
    metricName="recallByLabel",
)

lr = LogisticRegression(
    featuresCol="features",
    labelCol=target_column,
)
final_model = lr.fit(train_df)
preds = final_model.transform(test_df)
```

Report Model Performance Metrics on the Test Set

- Accuracy
- F1 Score
- Precision
- Recall

In [51]:
```
print(f"Model Accuracy: {acc_evaluator.evaluate(preds)}")
print(f"Model F1 Score: {f1_evaluator.evaluate(preds)}")
print(f"Model Precision: {precision_evaluator.evaluate(preds)}")
print(f"Model Recall: {recall_evaluator.evaluate(preds)}")
```

```
Model Accuracy: 0.9083333333333333
Model F1 Score: 0.9081739130434783
Model Precision: 0.8769230769230769
Model Recall: 0.95
```

Results

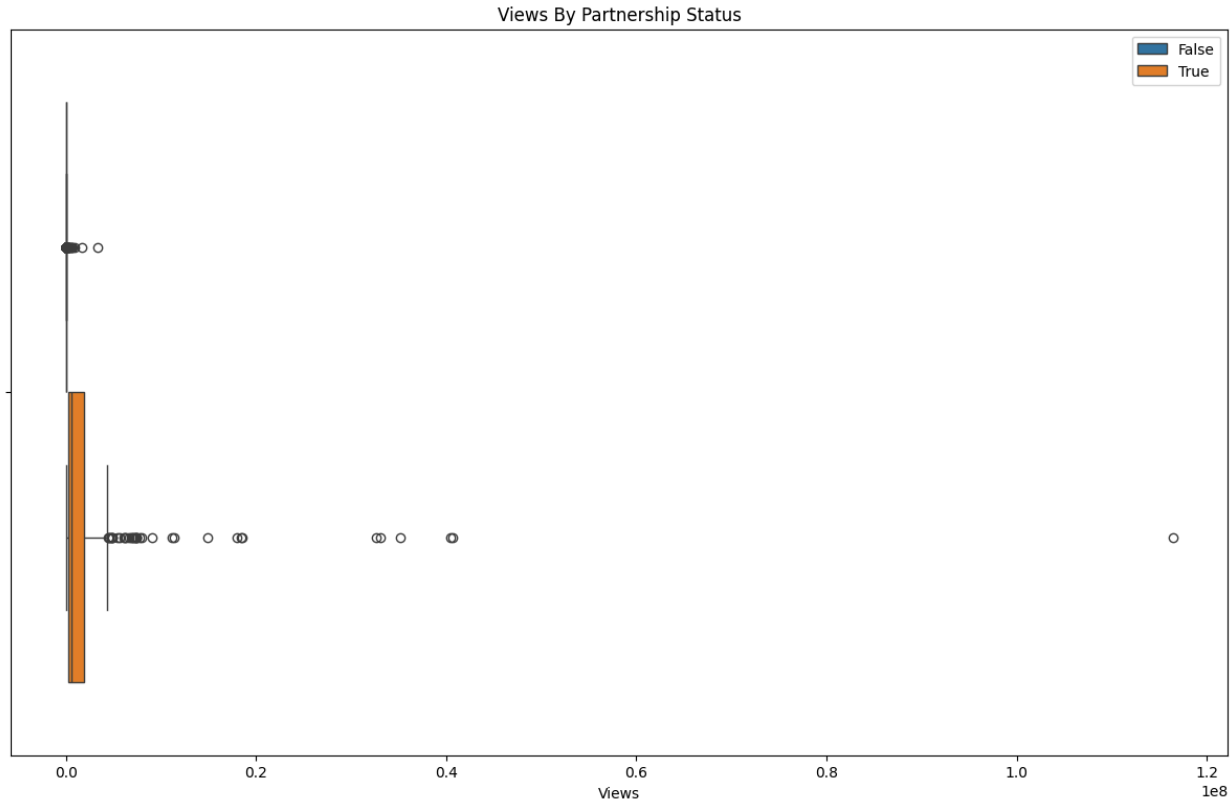
> The average metrics for Precision and Recall were a little higher for the k-folds cross-validation. The final model that was trained on the full training set outperformed the k-folds cross-validation method on Accuracy and the F1 score. This might be due to the small sample size once the class imbalance was handled.

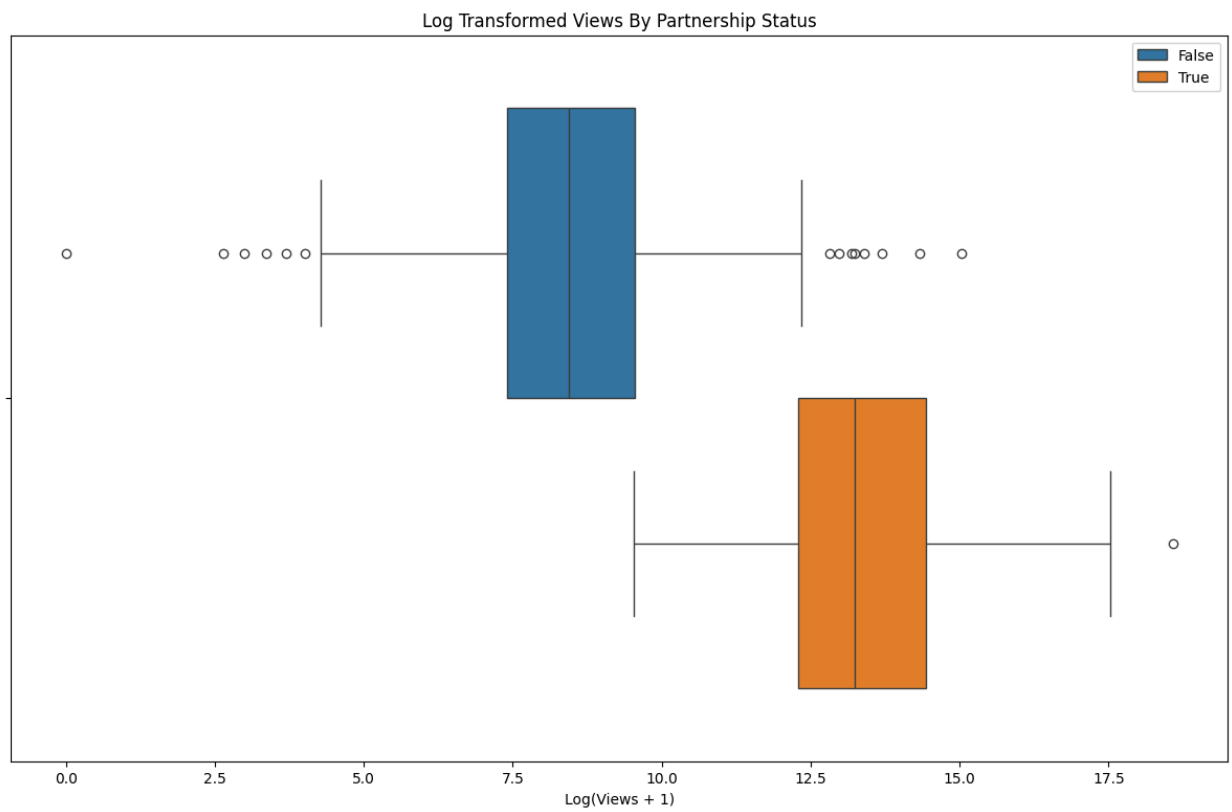
Extension Example For Views

> When looking at the numerical feature *views* there is a major skewness. In order to fix this I reran the model were a **Log Transformation** $log(x+1)$ was applied to the *views* feature.

Plots of the *Views* Numerical Feature with and without the log transformation

In [52]:
```
join_df = join_df.withColumn(
    "log_views", F.log(F.col("views") + 1))

numeric_boxplots(
    df=join_df,
    feature="views",
    target=target_column,
    title="Views By Partnership Status",
    x_label="Views",
)
numeric_boxplots(
    df=join_df,
    feature="log_views",
    target=target_column,
    title="Log Transformed Views By Partnership Status",
    x_label="Log(Views + 1)",
)
```





Adding the Transformed Column and Rebuilding the Features

In [53]:
```
log_train_df = train_df.withColumn(
    "log_views", F.log(F.col("views") + 1))
log_test_df = test_df.withColumn(
    "log_views", F.log(F.col("views") + 1))
```

```
feature_columns = [
    "degree",
    "betweenness",
    "closeness",
    "eigenvector",
    "pagerank",
    "days",
    "mature",
    "log_views",
]

assembler = VectorAssembler(
    inputCols=feature_columns,
    outputCol="log_features",
)
log_train_df = assembler.transform(log_train_df)
log_test_df = assembler.transform(log_test_df)
```

Rerunning the Final Model with New Features

In [54]:
```
log_lr = LogisticRegression(
    featuresCol="log_features",
    labelCol=target_column,
)
log_test_model = log_lr.fit(log_train_df)
log_preds = log_test_model.transform(log_test_df)

print(f"Model Accuracy: {acc_evaluator.evaluate(log_preds)}")
print(f"Model F1 Score: {f1_evaluator.evaluate(log_preds)}")
print(f"Model Precision: {precision_evaluator.evaluate(log_preds)}")
print(f"Model Recall: {recall_evaluator.evaluate(log_preds)}")
```

```
Model Accuracy: 0.9666666666666667
Model F1 Score: 0.9666574048346763
Model Precision: 0.9827586206896551
Model Recall: 0.95
```

Extra Metrics for the Paper

Clustering Coefficient and Connect Components

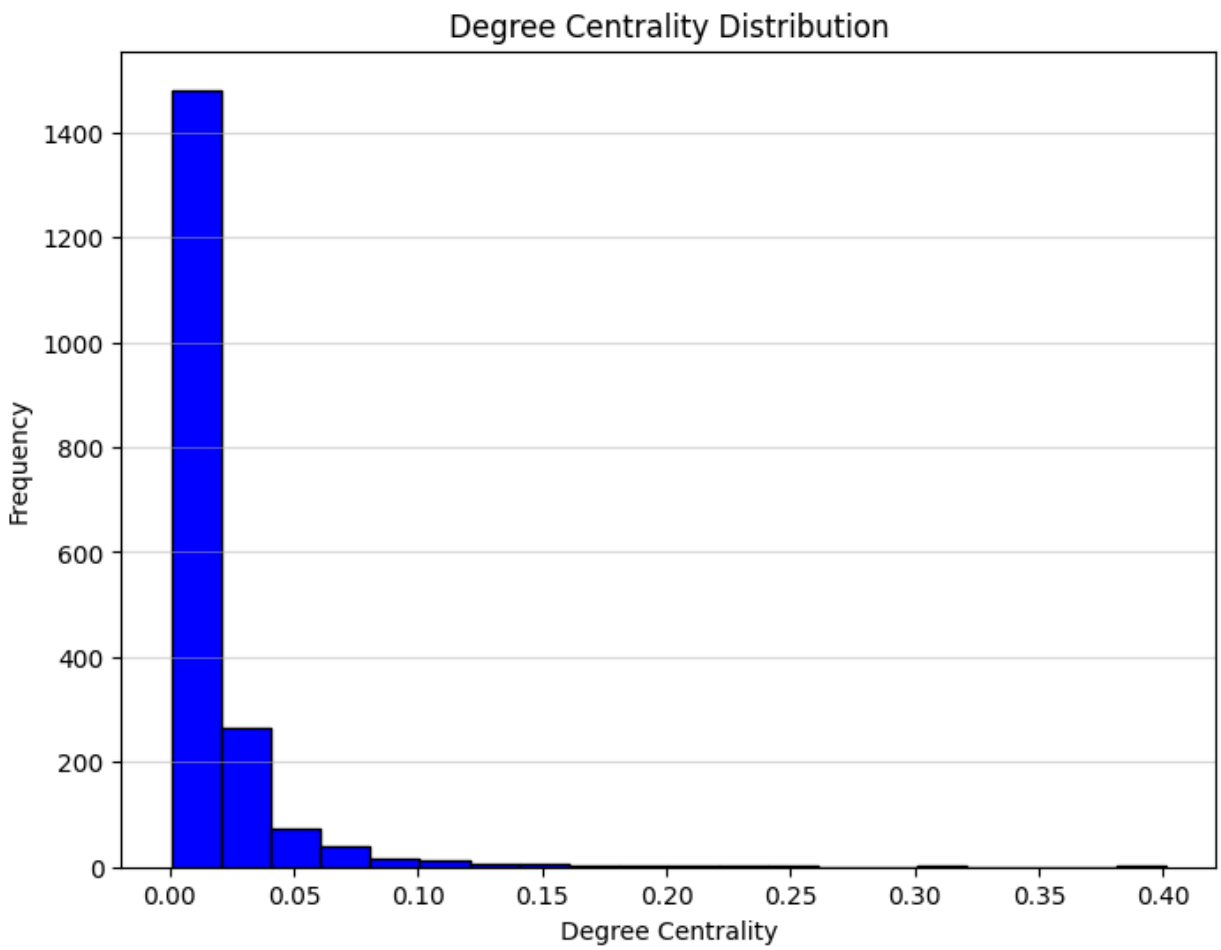
In [55]:
```
cc = nx.average_clustering(G)
print(f"Average Clustering Coefficient: {cc}")

components = nx.number_connected_components(G)
print(f"Number of Connected Components: {components}")
```

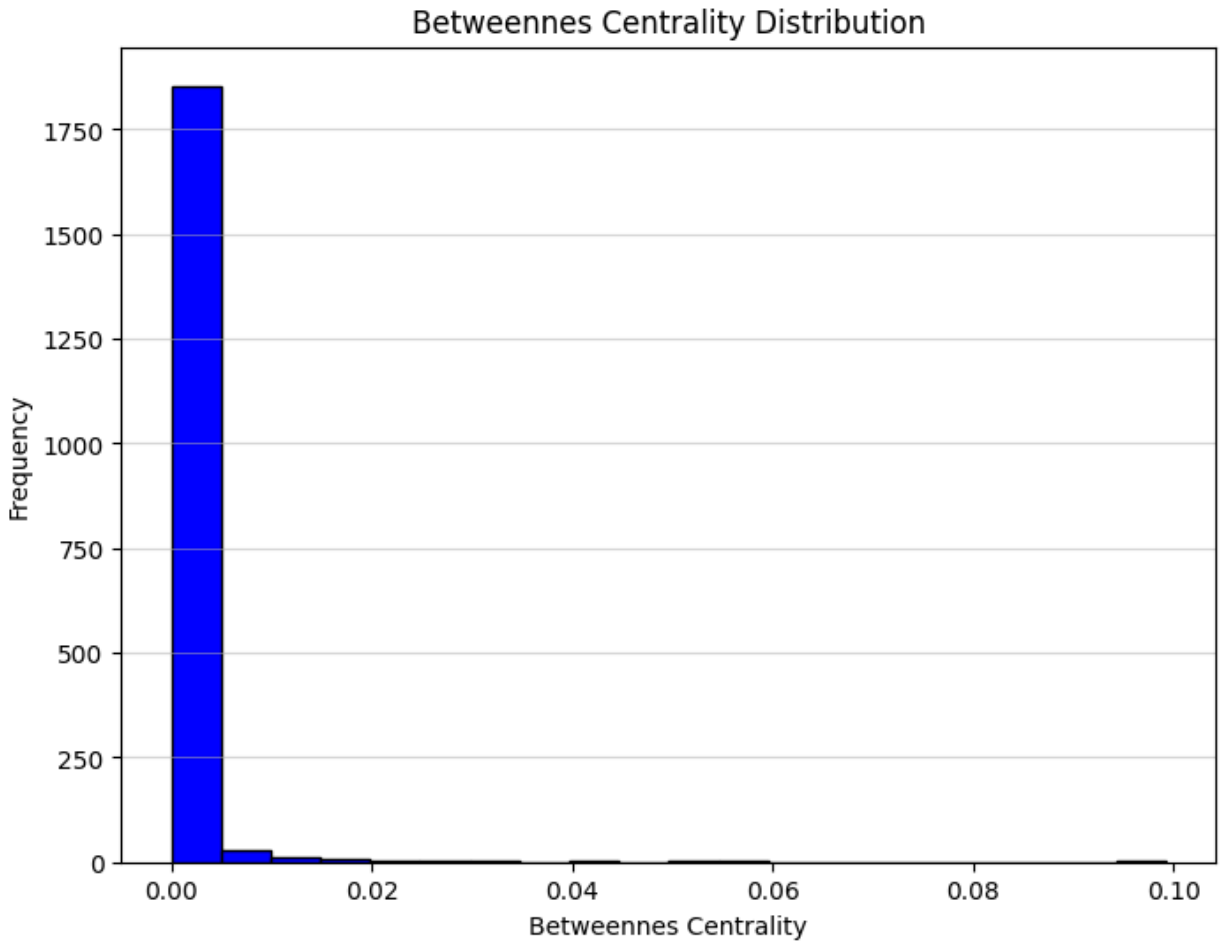
```
Average Clustering Coefficient: 0.31989527466935147
Number of Connected Components: 1
```

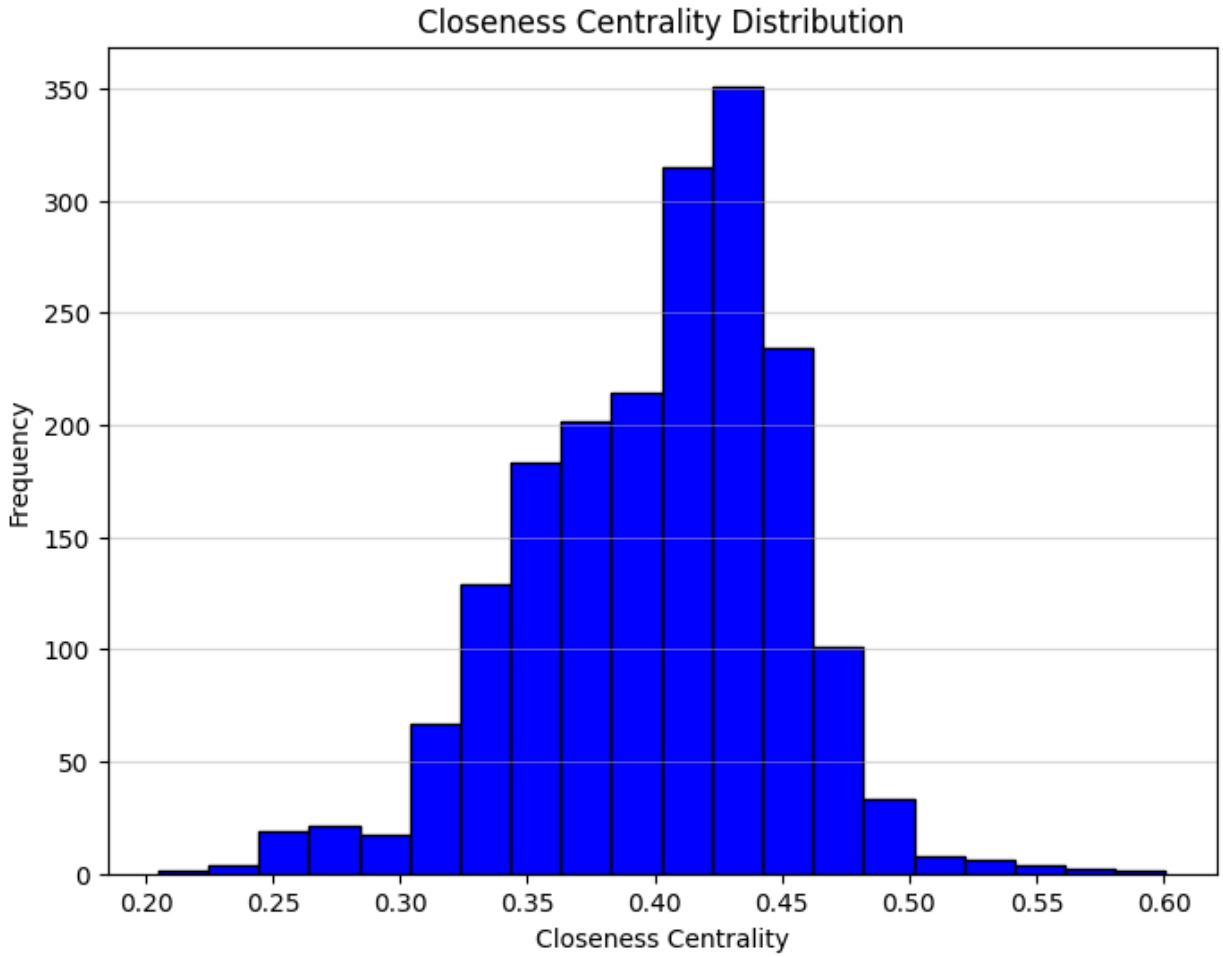
Histograms of Centralities

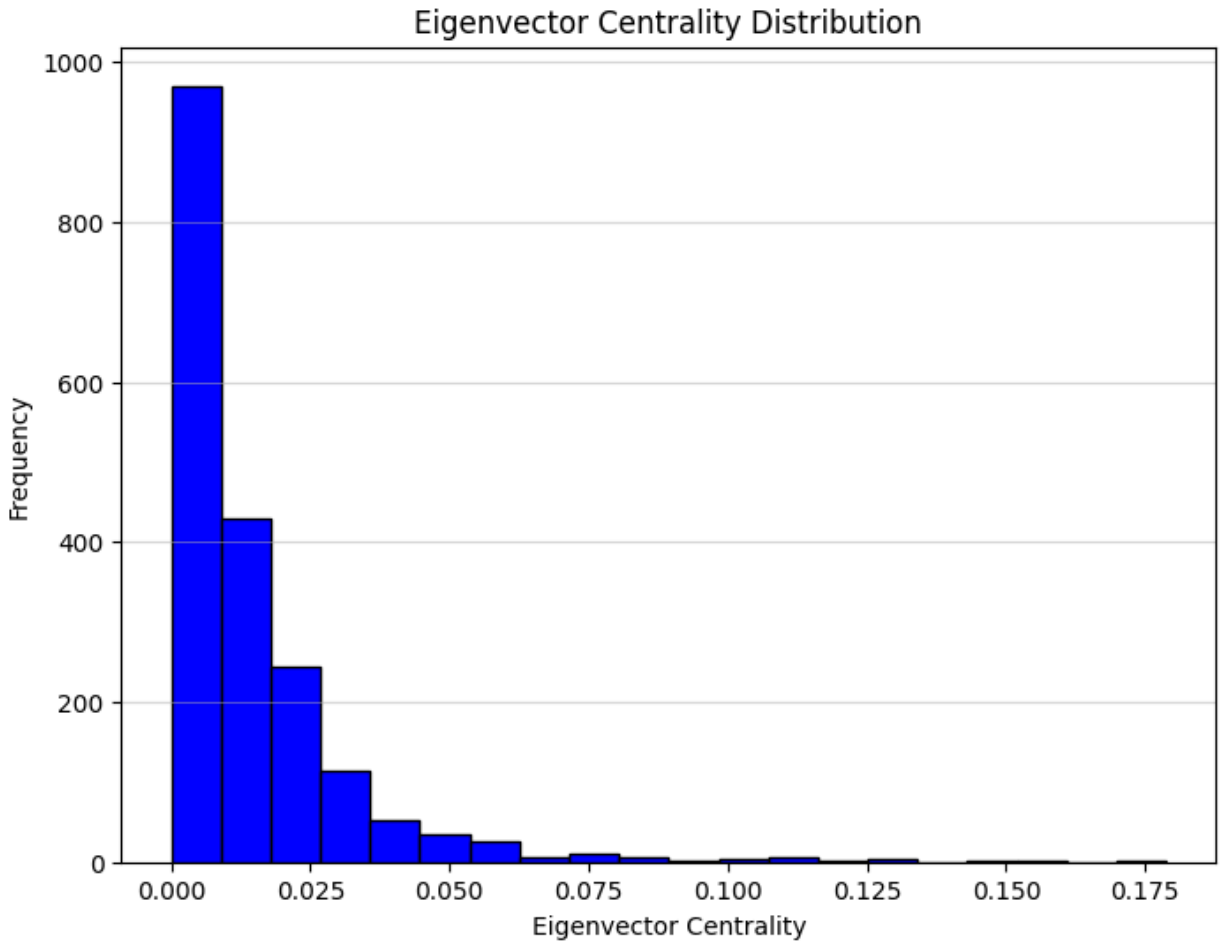
In [56]:
```
cent_list = [
    {"name": "degree", "val": list(degree.values())},
    {"name": "betweennes", "val": list(betweenness.values())},
    {"name": "closeness", "val": list(closeness.values())},
    {"name": "eigenvector", "val": list(eigenvector.values())},
    # {"name": "PageRank": pagerank},
]
```

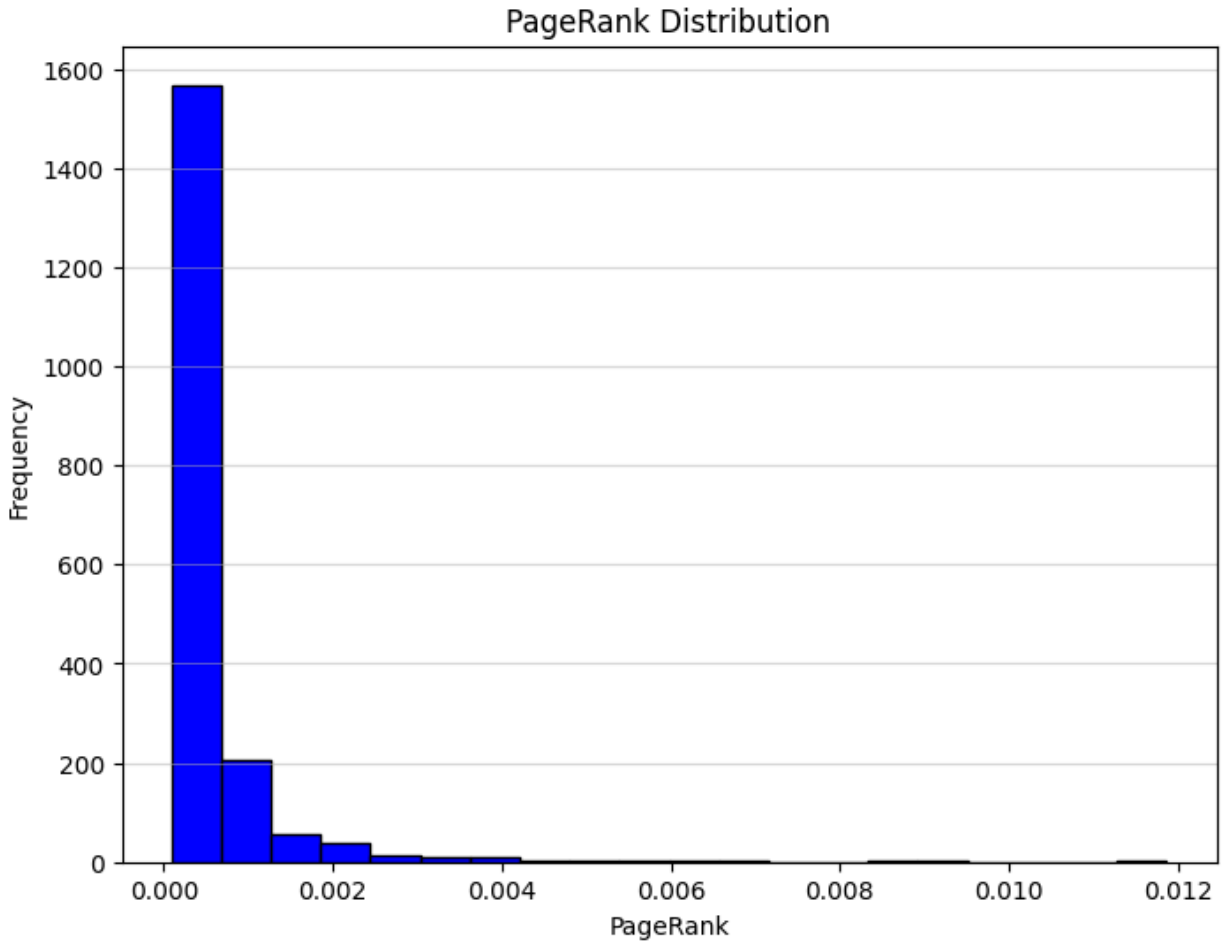
```python
for cent in cent_list:
    name = cent.get("name", "").title()
    vals = cent.get("val", [])
    plt.figure(figsize=(8, 6))
    plt.hist(vals, bins=20, color='blue', edgecolor='black')
    plt.title(f"{name} Centrality Distribution")
    plt.xlabel(f"{name} Centrality")
    plt.ylabel("Frequency")
    plt.grid(axis='y', alpha=0.5)
    plt.show()

plt.figure(figsize=(8, 6))
plt.hist(list(pagerank.values()), bins=20, color='blue', edgecolor='black')
plt.title(f"PageRank Distribution")
plt.xlabel(f"PageRank")
plt.ylabel("Frequency")
plt.grid(axis='y', alpha=0.5)
plt.show()
```




Betweennes Centrality Distribution
Frequency
1750
1500
1250
1000
750
500
250
0
0.00
0.02
0.04
0.06
0.08
0.10
Betweennes Centrality


Closeness Centrality Distribution
Frequency
350
300
250
200
150
100
50
0
0.20
0.25
0.30
0.35
0.40
0.45
0.50
0.55
0.60
Closeness Centrality


Eigenvector Centrality Distribution
Frequency
1000
800
600
400
200
0
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
Eigenvector Centrality



End of Project